

Seasonality, Pandemic-Period Shifts, and Hidden Markov Regimes in Monthly Malaria Counts from Three Ghanaian Regions, 2016–2023

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Abstract

Routine health information systems are increasingly used for malaria surveillance, but monthly facility counts may reflect seasonality, long-term trends, reporting changes, care-seeking behavior, and diagnostic practices. This study examined whether hidden Markov models (HMMs) identify distinct latent regimes in monthly malaria case counts from the Ashanti, Eastern, and Volta Regions of Ghana between January 2016 and December 2023. Region-specific Gaussian HMMs fitted to log-transformed counts identified lower- and higher-count states. These models were compared with Gaussian regression models incorporating seasonal terms, temporal trend, a COVID-period indicator, and first-order autoregressive (AR(1)) errors. The seasonal AR(1) model produced the lowest Akaike information criterion (AIC) and Bayesian information criterion (BIC) values in all three regions. Within the HMM family, AIC favored three states, whereas BIC favored two. A two-state HMM was also fitted to innovations from the seasonal AR(1) model. After seasonal and serial dependence were removed, the concentration of decoded states by calendar month declined approximately six- to eight-fold, and neither AIC nor BIC favored an additional two-state structure. Seasonal peaks occurred between July and September. Estimated COVID-period effects were negative but imprecise after accounting for serial dependence. Overall, the states identified in the unadjusted monthly counts mainly reflected recurring seasonality and month-to-month dependence rather than distinct underlying malaria-burden regimes.

Keywords: Malaria, Ghana, hidden Markov model, routine surveillance, seasonality, pandemic-period shifts, model comparison

1. Introduction

Malaria surveillance increasingly depends on routine health information systems. Facility-based reporting can provide monthly or even weekly information that is not available from periodic household surveys, and this makes routine data attractive for outbreak detection, resource allocation, intervention timing, and monitoring of changing malaria burden. The same feature that makes these systems valuable also makes them difficult to model. Monthly malaria counts are not direct observations of latent transmission intensity. They are a product of infection risk, care seeking, diagnostic availability, reporting completeness, health-system organization, population at risk, and seasonal environmental conditions. In endemic settings, a high-count month may therefore represent a genuine increase in transmission, an expected point in the annual cycle, a reporting change, a change in attendance, or some combination of these mechanisms.

Ghana is an appropriate setting for this modelling problem because malaria burden is spatially and temporally heterogeneous. Prior work using Ghana routine health facility data has shown that malaria morbidity differs by place and time, and that national summaries can conceal substantial subnational variation (Awine et al., 2018). At the same time, a routine monthly regional series is not the same object as a designed incidence study. The numerator is reported cases, while denominators such as population at risk, facility catchment size, and reporting completeness may be unavailable. A model that appears to discover “high burden states” can therefore be scientifically useful only if its states are separated from obvious calendar effects and known reporting-period shifts.

Hidden Markov models (HMMs) offer an appealing way to summarize such longitudinal surveillance data. In an HMM, observations are generated from a latent state process, and the state process follows a Markov transition structure (Rabiner, 1989; Zucchini et al., 2016). For monthly malaria counts, a two-state HMM may be read as a lower- and higher-count-regime model. The transition matrix then gives expected state persistence, and the decoded state sequence gives an interpretable temporal classification. This interpretation is attractive for surveillance because it appears to translate noisy monthly counts into regimes that can be tracked, compared, and potentially acted on.

The difficulty is that this interpretation can be too strong. A stationary HMM without seasonal covariates must use latent states to explain all recurring changes in the observed series. If the dominant pattern is the annual malaria cycle, the HMM may decode wet-season months as a higher state and dry-season months as a lower state. The model can look coherent, and its states can even be well separated on the log-count scale, while still failing to identify seasonally independent burden regimes. A similar concern applies to the COVID-19 period. The 2020–2023 window may include changes in care seeking, facility use, program delivery, diagnostics, or reporting. If these changes are not represented explicitly, a time-homogeneous HMM may use latent states to absorb both disease dynamics and reporting-period shifts.

We test whether simple HMM regimes add information beyond explicit seasonal, trend, and COVID-period structure in monthly malaria counts from three Ghanaian regions. The three-region monthly dataset cannot support a national malaria-regime map. Instead, the analysis examines whether decoded states retain support after model checking and can be distinguished from predictable seasonal structure.

We compare region-specific Gaussian HMMs for monthly malaria counts in Ashanti, Eastern, and Volta from January 2016 through December 2023 with a one-state Gaussian model and seasonal/trend/COVID-period Gaussian models on the same log-count scale. An innovation diagnostic removes the fitted seasonal mean and AR(1) dependence before testing whether a two-state HMM improves on a single Gaussian. The assessment includes AIC–BIC disagreement, decoded-state seasonality, COVID-period state occupancy, emission overlap, posterior decoding confidence, block-bootstrap uncertainty, one-step probability integral transform (PIT) summaries, and estimates from the seasonal AR(1) model. The results show that the stationary HMM states fitted to raw counts are explained by seasonality and serial dependence.

2. Related work

2.1 Routine malaria surveillance and Ghana

Routine health facility data have become central to malaria monitoring because they provide frequent observations at operational scales. For malaria programs, such data can support subnational trend monitoring, seasonally timed intervention planning, and detection of unusual increases in reported cases. The World Malaria Report annexes and related surveillance resources document national and regional malaria indicators, but they do not remove the need for careful interpretation of routine facility counts (World Health Organization, 2025). Facility counts can change when transmission changes, but they can also change when reporting systems, access to care, testing behavior, or denominator populations change.

For Ghana, Awine et al. (2018) show spatio-temporal heterogeneity in routine malaria morbidity and the value of subnational analysis. Morbidity seasonality follows ecological zone and rainfall: the Guinea savannah has one wet season and one annual morbidity peak, whereas transitional forest and coastal savannah zones have two wet seasons and bimodal morbidity. Ashanti, Eastern, and Volta span forest, transitional, and coastal-savannah conditions, so their monthly counts are expected to contain a strong annual signal. Spatio-temporal variation in routine counts does not necessarily represent variation in true incidence because care seeking, testing, and reporting also affect the observations.

More recent Ghanaian work reinforces both points. Antwi et al. (2026) fit a Bayesian hierarchical spatio-temporal model to climate-linked malaria data and report that reported malaria peaks roughly one to two months after heavy rainfall, with significant spatial dependence and temporal correlation, and that overlooking such temporal structure can obscure dynamics and bias model-based inference. Survey-based studies provide complementary burden context: Tetteh et al. (2023) use the Ghana Demographic and Health and Malaria Indicator Surveys to quantify infection prevalence and the protective effect of bed-net distribution among children aged 6–59 months, and Klu et al. (2026) link household water, sanitation, and hygiene conditions to malaria prevalence in the 2022 survey, noting that transmission in Ghana occurs year round but is sustained by environmental and socio-economic conditions. The WHO Ghana country profile situates these regional series within a high-burden national setting (World Health Organization, 2024). None of these sources remove the need to separate calendar effects from regime structure in a routine monthly series; together they make the separation more pressing, because the expected seasonal signal is large and well documented.

Unlike broad national spatio-temporal mapping studies, our analysis uses a public dataset containing monthly malaria, HIV, and tuberculosis counts for three regions. This spatial coverage precludes national generalization but permits a focused evaluation of latent-regime claims in routine monthly surveillance series. The analysis is therefore methodological rather than a comprehensive disease-atlas study.

2.2 Spatial and spatio-temporal surveillance methods

Spatio-temporal surveillance methods must address both space-time structure and the observation process. Lan and Delmelle (2023) review space-time cluster detection techniques for infectious diseases, showing the breadth of methods used to identify unusual geographic and temporal concentration. Their review is relevant here because latent states and cluster-like labels can influence surveillance decisions; if such labels are driven by predictable seasonal structure, their operational meaning changes.

Li et al. (2020) focus on a related problem: passive surveillance data can be biased, and model-based integration or correction may be needed when passive data are combined with active surveillance. Although the current paper does not integrate active and passive data, the same principle applies. Routine monthly counts are useful observations, but the observation model matters. A latent state estimated from passive counts may reflect reporting intensity or attendance as much as underlying transmission.

More recent spatio-temporal infectious disease work also illustrates the value of explicit temporal structure. Adeoye et al. (2026) present Bayesian spatio-temporal modelling for outbreak detection, and their approach reflects a broader movement toward models that separate latent risk dynamics from expected variation and reporting noise. The present study is simpler by design, but it follows the same diagnostic logic: before interpreting a state label as an epidemiological regime, compare it with more transparent explanations for temporal variation.

2.3 Latent states, temporal clustering, and malaria

Malaria temporal clustering studies show how time-series summaries can inform surveillance and intervention planning. Legendre et al. (2023) use temporal dynamic clustering for malaria surveillance, emphasizing that temporal patterns are relevant to programmatic decisions. Such work supports the practical value of identifying recurring temporal profiles. It also highlights the distinction between temporal classification and causal interpretation. A useful cluster or state can support planning without being a direct estimate of an unobserved biological regime.

HMMs are one member of this wider class of state-based temporal models. They have a long history in time-series classification and latent-state inference (Rabiner, 1989; Zucchini et al., 2016). Early infectious-disease applications include the surveillance model of Le Strat and Carrat (1999), which allowed for trend and seasonality and illustrated Poisson emissions using poliomyelitis counts, and the automated influenza-epidemic detector of Rath et al. (2003). Count time series can be modelled directly with Poisson or negative-binomial emissions (Zucchini et al., 2016), although continuous approximations on log-transformed counts remain common in practice when volumes are large. The standard time-homogeneous HMM assumes that transition probabilities do not vary with calendar month unless covariates or non-homogeneous transition structures are introduced. When seasonality is strong, that assumption can make state interpretation fragile.

Model-selection criteria do not fully solve this problem. AIC rewards fit with a smaller penalty, while BIC uses a stronger penalty that increases with sample size (Akaike 1974; Schwarz, 1978). In small monthly surveillance series, AIC and BIC may recommend different HMM state counts. Even if an HMM is preferred within the HMM family, it may still be inferior to a regression baseline with explicit seasonal terms. This study therefore treats model comparison as a broader benchmarking exercise, not as a choice only among two-state and three-state HMMs.

3. Data

3.1 Source and provenance

The primary dataset is *Indicators Data*, Version 1, a public secondary dataset published by Issah Seidu on Mendeley Data on 11 May 2026 under a CC BY 4.0 license (Seidu, 2026). The repository describes the observations as monthly reported malaria, HIV, and tuberculosis counts for Ashanti, Eastern, and Volta Regions extracted from Ghana Health Service DHIMS II. We did not perform the original extraction or have access to the underlying GHS database. Accordingly, we treat the deposited file as a secondary dataset rather than an official GHS release.

The analytic panel contains 96 monthly observations per region, covering January 2016 through December 2023, for 288 region-month observations overall, with no missing months. Malaria counts are the outcome; the HIV and tuberculosis counts are not used here, as the question concerns malaria seasonality and latent count regimes.

The series pass basic face-validity checks. The regional ordering of reported volume (Ashanti > Eastern > Volta) matches these regions' relative population and known malaria burden; the monthly counts are strictly positive with no duplicated periods; and the fitted seasonal peaks (July–September) coincide in timing and ecological zone with the wet-season transmission pattern reported for Ghana (Awine et al., 2018; Antwi et al., 2026). These checks do not certify reporting completeness, but they confirm that the series are internally consistent and consistent in scale and timing with external knowledge of Ghanaian malaria.

3.2 Variables and transformations

Let Y_{rt} denote the reported malaria case count for region r in month t . The primary model outcome is $X_{rt} = \log(Y_{rt})$. The logarithmic transformation is used because the regional count levels differ substantially, the Gaussian HMM is fitted on a continuous scale, and multiplicative differences in counts are more interpretable for monthly surveillance data than additive differences on the raw-count scale. All reported count-scale state summaries are back-transformed with lognormal interpretation: $\exp(\mu)$ is a median and $\exp(\mu + \sigma^2/2)$ is a mean when $X \sim N(\mu, \sigma^2)$.

Because monthly regional counts are large, overdispersed, and lack reliable denominators or exposure offsets, we model log counts with Gaussian emissions as a descriptive surveillance-scale model rather than as an individual-count generative model. Poisson or negative-binomial HMMs would be natural extensions if denominators, reporting completeness, or exposure offsets were available.

Calendar month is represented with the first two annual sine/cosine harmonics, where m is month of year. This gives the seasonal baseline flexibility for one and two annual peaks without estimating a separate parameter for every month. A centered linear time trend represents gradual changes over the eight-year window. The supplied COVID-period indicator is included as a binary reporting-period covariate; it equals 1 from April 2020 through December 2023 (capturing 45 months of the pandemic period) and 0 otherwise. It is interpreted descriptively, not causally.

3.3 Scope and limitations of the data

The dataset is useful because it provides a clean monthly regional panel, but it is not sufficient for incidence modelling. It lacks population denominators, facility-level reporting completeness, diagnostic testing volumes, catchment population changes, and information on changes in care-seeking or facility attendance. The paper therefore uses the phrase “case-count regime” rather than “incidence regime.” The distinction is substantive: a higher reported count state may arise from higher incidence, but it may also arise from improved reporting, more facility attendance, or a larger population at risk.

The spatial scope is also limited. Ashanti, Eastern, and Volta are important Ghanaian regions, but they do not constitute a national spatial sample. The regional comparison is therefore descriptive and diagnostic. The results can show how HMM interpretations behave in three routine malaria series, but they cannot support claims about all Ghanaian regions or district-level heterogeneity. A national spatio-temporal model would require broader regional or district coverage and ideally denominators or reporting-quality measures.

4. Methods

4.1 Analytic design

The analysis is organized around model checking. For each region, four model classes are fitted to the same log-monthly malaria-count series: a one-state Gaussian model, a seasonal/trend/COVID-period Gaussian baseline, a two-state Gaussian HMM, and a three-state Gaussian HMM. The one-state Gaussian model provides a minimal reference for unstructured variation. The seasonal/trend/COVID-period model provides a transparent benchmark for calendar and reporting-period structure. The two-state and three-state HMMs provide latent-regime alternatives.

This comparison answers whether an HMM with no calendar terms beats a regression with calendar terms, but the model families do not share covariates. We therefore add an innovation diagnostic (Section 4.5) that removes the seasonal, trend, COVID-period, and AR(1) structure before testing whether a two-state HMM improves on a single Gaussian. The first comparison evaluates the raw series, while the second tests for remaining state structure after the observed temporal pattern has been modelled.

All comparisons are performed separately by region. This choice avoids imposing spatial dependence that the three-region dataset cannot estimate credibly. It also makes the interpretation of each time series transparent. The cost is that spatial borrowing and shared regional dynamics are not modelled. That cost is acceptable for the diagnostic question because

the main concern is whether each region-specific HMM is outperforming an explicit seasonal benchmark.

4.2 Gaussian baseline models

The one-state Gaussian model assumes

$$X_{rt} \sim N(\mu_r, \sigma_r^2),$$

independently over months conditional on the region-specific parameters. It has two parameters per region and serves as a minimal distributional reference.

The seasonal/trend/COVID-period Gaussian baseline assumes

$$X_{rt} = \alpha_r + \beta_r c_t + \gamma_{1r} \sin(2\pi m_t/12) + \gamma_{2r} \cos(2\pi m_t/12) + \gamma_{3r} \sin(4\pi m_t/12) + \gamma_{4r} \cos(4\pi m_t/12) + \delta_r C_t + \epsilon_{rt},$$

where c_t is centered time, m_t is month of year, C_t is the COVID-period indicator, and $\epsilon_{rt} \sim N(0, \sigma_r^2)$. The model has eight estimated quantities per region: seven regression coefficients and one residual variance. It is not the best possible model for malaria counts but a simple benchmark for the most obvious non-HMM temporal structure.

The stronger benchmark allows AR(1) residual errors, $\epsilon_{rt} = \phi_r \epsilon_{r,t-1} + \eta_{rt}$, where $\eta_{rt} \sim N(0, \sigma_{\eta,r}^2)$ and $|\phi_r| < 1$. It is estimated iteratively using the Prais–Winsten transformation and has nine parameters per region: seven regression coefficients, one residual variance, and ϕ_r . Coefficient standard errors are calculated from the transformed design matrix and innovation variance. The first-harmonic amplitude is $A_r = \sqrt{\gamma_{1r}^2 + \gamma_{2r}^2}$; its peak month maximizes $\gamma_{1r} \sin(2\pi m/12) + \gamma_{2r} \cos(2\pi m/12)$; and the multiplicative COVID-period effect is $\exp(\delta_r)$, with a Wald 95% interval $\exp(\hat{\delta}_r \pm 1.96 \text{ se}(\hat{\delta}_r))$. These reported estimates and intervals come from the AR(1) model selected in the cross-family comparison.

4.3 Hidden Markov models

For each region, Gaussian HMMs are fitted to X_{rt} . Let $S_{rt} \in \{1, \dots, K\}$ be the latent state for region r at month t , with $K = 2$ or $K = 3$. The state process follows

$$\Pr(S_{rt} = j | S_{r,t-1} = i) = p_{ijr},$$

and the observation distribution is

$$X_{rt} | S_{rt} = k \sim N(\mu_{kr}, \sigma_{kr}^2).$$

States are ordered after fitting by increasing emission mean so that the first state is the lower count state and the final state is the higher count state. The two-state HMM therefore yields lower and higher count summaries. For the three-state HMM, model selection is reported but the main interpretive diagnostics focus on the two-state model because BIC selects two states within the HMM family.

HMM likelihoods are evaluated with forward recursions, and most likely state sequences are decoded with the Viterbi algorithm (Rabiner 1989; Zucchini et al. 2016). The expected duration

of state i is summarized as $1/(1 - p_{ii})$ months. This duration formula is interpretable only under the time-homogeneous Markov assumption, which is one reason the seasonality diagnostics matter.

4.4 Model comparison

Model comparison uses AIC and BIC:

$$\text{AIC} = -2\ell + 2q, \quad \text{BIC} = -2\ell + q\log(n),$$

where ℓ is the maximized log likelihood, q is the number of estimated parameters, and $n = 96$ months for each region. AIC and BIC answer related but not identical questions. AIC gives a lighter penalty to additional parameters and often favors more flexible models (Akaike 1974). BIC penalizes model complexity more strongly at this sample size (Schwarz 1978). Both criteria are reported because disagreement between them is itself a warning in a short monthly series. The series are also strongly autocorrelated (the PIT lag-1 autocorrelations below reach 0.56), so the effective sample size is well below 96; the BIC penalty $q\log n$ is then, if anything, lenient toward the more heavily parameterized models.

The central comparison is not only two-state versus three-state HMM. A model can be the best HMM and still fail to outperform a simple seasonal regression. The analysis therefore compares all four model classes together, so that an HMM-only selection is not mistaken for evidence that a latent-regime explanation is necessary.

4.5 Innovation state-structure diagnostic

The seasonal baseline contains calendar covariates that are absent from the raw-count HMMs. To examine the remaining structure, we use the innovations from the fitted seasonal AR(1) model. The first innovation is scaled by $\sqrt{1 - \phi_r^2}$, and subsequent innovations are $\eta_{rt} = e_{rt} - \phi_r e_{r,t-1}$, where $e_{rt} = X_{rt} - \hat{X}_{rt}$. A single Gaussian and a two-state Gaussian HMM are fitted to these innovations. Lower AIC or BIC for the two-state model would indicate structure not captured by the seasonal mean or first-order serial dependence.

For the raw-count HMM and innovation HMM, we also compute the share of observations decoded as the higher state within each calendar month and summarize the twelve shares by their standard deviation. High dispersion indicates that the state assignments follow the annual cycle. The change after AR(1) filtering shows how much calendar concentration remains once seasonality and serial dependence have been represented directly.

4.6 Interpretive decision rules

The analysis applies conservative decision rules before treating an HMM state as an epidemiologically meaningful burden regime. First, the HMM must be compared with transparent non-HMM baselines fitted to the same transformed outcome. If a seasonal baseline is preferred, the HMM state labels are interpreted as exploratory classifications rather than evidence of an additional latent process. Second, the HMM-only state-count choice must be reported with both AIC and BIC. If the criteria disagree, the state-count interpretation is considered unstable. Third, decoded state assignments must be checked against calendar month. A state that is concentrated in predictable months may still be useful operationally, but it should

not be described as seasonally independent. Fourth, state separation and uncertainty must be reported on the count scale. Large differences in back-transformed means do not remove the need to report overlap, posterior uncertainty, and duration uncertainty.

These rules are stricter than a routine HMM fitting exercise, which is appropriate for passive reported counts with limited covariate information. A latent-state label is useful for decision making only if the reader knows whether it captures unexpected burden, expected seasonality, reporting-period shifts, or a mixture of these.

4.7 State-separation and decoding diagnostics

For the selected two-state HMM summaries, emission separation is evaluated with three quantities. First, the lower and higher state medians and lognormal means are reported on the original count scale. Second, the log-mean difference is standardized by a pooled emission standard deviation. Third, the overlap coefficient between the two fitted normal emission densities is computed numerically. An overlap coefficient near zero would indicate near-complete separation; larger values indicate that observations can plausibly arise from either state.

Posterior decoding confidence is summarized by the mean posterior probability of the decoded state and by the share of months whose decoded-state posterior probability is below 0.80. These quantities do not validate the epidemiological meaning of a state, but they show how uncertain the state assignment is under the fitted HMM.

4.8 Seasonality, COVID-period, and residual diagnostics

Decoded seasonality is measured by the share of months assigned to the higher HMM state for each calendar month. If higher-state assignments concentrate in a small set of predictable months, then the HMM is acting partly as a seasonal classifier. COVID-period diagnostics compare higher-state occupancy before and during the supplied COVID-period indicator. These comparisons are descriptive because the dataset does not identify the causal effect of the pandemic or separate policy, reporting, and care-seeking mechanisms.

Goodness of fit is screened with one-step probability integral transform (PIT) values. PIT diagnostics are widely used to assess whether predictive distributions are calibrated; randomized quantile residuals and related PIT ideas are especially relevant for non-Gaussian or discrete responses (Dunn and Smyth 1996). These are *in-sample* one-step-ahead filtered values: the model is fitted to the full series, and at each month the one-step predictive distribution is formed from the filtered state distribution propagated through the transition matrix. They are a calibration screen, not a held-out forecasting evaluation, and the autocorrelation result below should be read accordingly. The analysis reports the PIT mean, a Kolmogorov–Smirnov distance from uniformity, and lag-1 autocorrelation in the PIT sequence. PIT means near 0.5, and small KS distances are reassuring, but substantial lag-1 autocorrelation suggests residual temporal dependence not captured by the fitted model.

4.9 Uncertainty assessment

Uncertainty in selected HMM quantities is assessed with 50 circular block-bootstrap replicates using 12-month blocks. The block length preserves within-year dependence better than an independent bootstrap and is aligned with the annual cycle. For each replicate, the two-state HMM is refitted and state summaries are re-estimated. The reported intervals are empirical 2.5th

and 97.5th percentiles across bootstrap replicates. With 50 replicates, the intervals should be read as approximate stability checks rather than high-precision confidence intervals.

5. Results

5.1 Descriptive monthly malaria counts

The three regions differ substantially in reported monthly case volume (Table 1). Ashanti has the largest total count over the observation window, with 7.29 million reported malaria cases and a mean of 75,958 cases per month. Eastern follows with 6.03 million total cases and a mean of 62,784 per month. Volta has 2.53 million reported cases and a mean of 26,326 per month. The regional differences justify fitting region-specific models rather than pooling the raw counts.

Table 1: Analytic monthly routine-surveillance panel by region

Region	First month	Last month	Months	Malaria total	Monthly mean	Monthly min	Monthly max
Ashanti	2016-01	2023-12	96	7,292,008	75,958.42	49,717	105,202
Eastern	2016-01	2023-12	96	6,027,238	62,783.73	35,884	94,439
Volta	2016-01	2023-12	96	2,527,342	26,326.48	9,977	42,120

All three series show recurring within-year variation. This is visible in the decoded time-series in Figure 1 and is reflected in the strong month-of-year patterns reported below. The descriptive pattern is already enough to make model criticism necessary: if a latent-state model is fitted without calendar terms, the states may represent expected seasonal fluctuation rather than unexpected regime switching.

Monthly malaria case counts and decoded two-state HMM regimes, 2016--2023

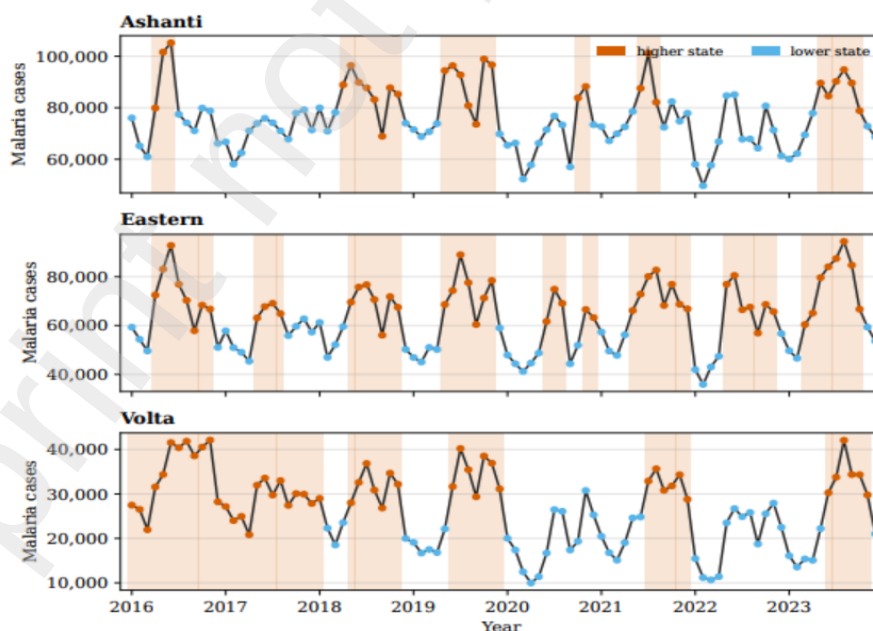


Figure 1: Monthly malaria case counts and decoded two-state HMM regimes for Ashanti, Eastern, and Volta, January 2016–December 2023. The figure shows the operational appeal of the HMM summary:

each month receives a lower or higher state label. The subsequent diagnostics evaluate whether these labels add information beyond seasonality and reporting-period structure.

5.2 Model comparison

The model comparison is clear-cut (Table 2). In every region, the seasonal baseline models outperform the HMMs, and the seasonal/trend/COVID-period Gaussian baseline with AR(1) residual errors has the lowest AIC and the lowest BIC overall. In Ashanti, the seasonal AR(1) baseline has BIC -135.163, and the standard seasonal baseline has BIC -116.881, compared with -80.651 for the one-state Gaussian model, -76.156 for the two-state HMM, and -66.131 for the three-state HMM. In Eastern, the seasonal AR(1) baseline has BIC -102.118, and the standard seasonal baseline has BIC -82.379, compared with -19.260, -25.578, and -13.643 for the one-state model, two-state HMM, and three-state HMM, respectively. In Volta, the seasonal AR(1) baseline has BIC -43.155, and the standard seasonal baseline has BIC 1.663, compared with 78.395, 53.020, and 60.661.

Table 2: Model comparison by region

Region	Model	Log likelihood	Parameters	AIC	BIC
Ashanti	One-state Gaussian	44.890	2	-85.780	-80.651
Ashanti	Seasonal/trend/COVID baseline	76.698	8	-137.396	-116.881
Ashanti	Seasonal/trend/COVID baseline with AR(1)	88.121	9	-158.242	-135.163
Ashanti	Two-state HMM	54.053	7	-94.106	-76.156
Ashanti	Three-state HMM	65.016	14	-102.031	-66.131
Eastern	One-state Gaussian	14.194	2	-24.388	-19.260
Eastern	Seasonal/trend/COVID baseline	59.447	8	-102.894	-82.379
Eastern	Seasonal/trend/COVID baseline with AR(1)	71.598	9	-125.197	-102.118
Eastern	Two-state HMM	28.764	7	-43.529	-25.578
Eastern	Three-state HMM	38.772	14	-49.544	-13.643
Volta	One-state Gaussian	-34.633	2	73.266	78.395
Volta	Seasonal/trend/COVID baseline	17.426	8	-18.852	1.663
Volta	Seasonal/trend/COVID baseline with AR(1)	42.117	9	-66.234	-43.155
Volta	Two-state HMM	-10.535	7	35.070	53.020
Volta	Three-state HMM	1.620	14	24.760	60.661

AIC gives the same cross-family ranking: the seasonal models are preferred in all three regions, with the seasonal AR(1) model performing best, so the result is not a BIC-only penalty effect. The HMMs improve log likelihood relative to the one-state model, but they are clearly outperformed by the seasonal baseline models. This confirms that the HMMs are largely capturing seasonal autocorrelation that is more parsimoniously and effectively represented by explicit seasonal terms and residual temporal dependence.

Within the HMM family, AIC and BIC disagree. AIC favors the three-state HMM for all three regions: the three-state HMM has lower AIC than the two-state HMM in Ashanti (-102.031 versus -94.106), Eastern (-49.544 versus -43.529), and Volta (24.760 versus 35.070). BIC favors the two-state HMM for all three regions because the additional seven parameters in the three-state HMM are not justified under the stronger penalty. The disagreement shows that the HMM state count is sensitive to the selection criterion in a 96-month series and reinforces the need to benchmark HMMs against non-HMM seasonal models. Figure 2 shows the BIC gap between each model and the seasonal baseline; every HMM and the one-state model sit well above the

baseline in all three regions, while the seasonal baseline with AR(1) residual errors lies significantly below it.

Model comparison: Δ BIC relative to the seasonal/trend/COVID baseline (lower is better)

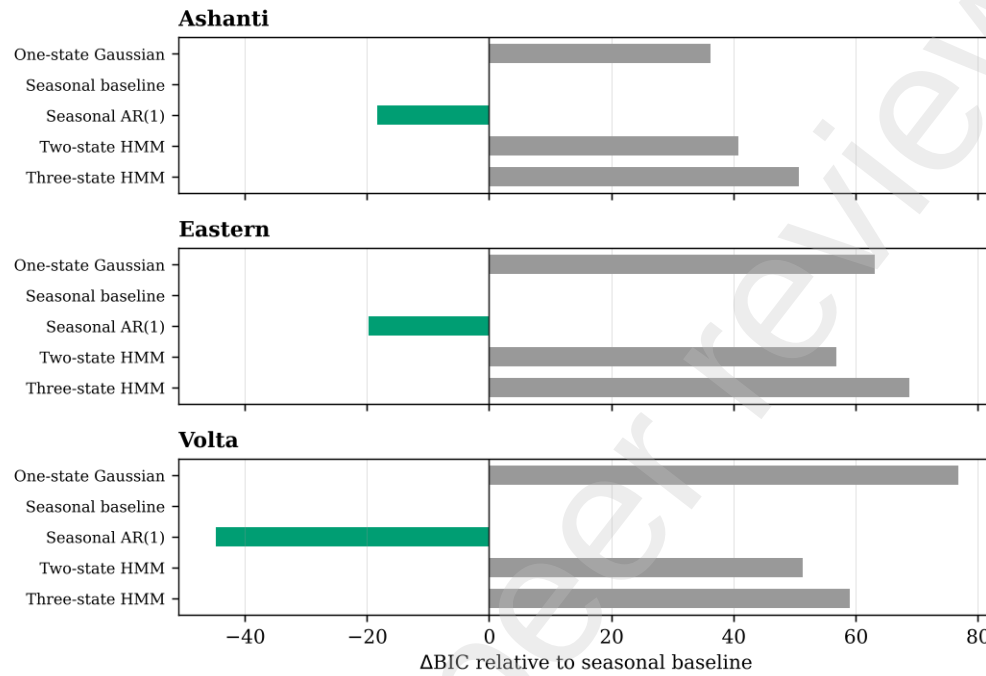


Figure 2: Model comparison by region, expressed as the BIC difference from the seasonal/trend/COVID-period baseline (the baseline is at zero; lower is better). In every region, the seasonal baseline with AR(1) residual errors achieves the lowest BIC overall, while the one-state Gaussian, two-state HMM, and three-state HMM are tens of BIC units worse than the seasonal models.

5.3 Characterizing the seasonal benchmark

The seasonal AR(1) estimates are reported in Table 3. The first-harmonic amplitude on the log scale rises from Ashanti (0.111, about a 12% multiplicative swing around the harmonic mean) through Eastern (0.208, about 23%) to Volta (0.341, about 41%). The fitted first-harmonic peak falls in July for Ashanti, August for Eastern, and September for Volta. This window agrees with the wet-season transmission timing documented for these ecological zones and the reported rainfall lag in Ghana (Awine et al. 2018; Antwi et al. 2026). The model explains 48%, 61%, and 66% of the log-count variance. The estimated COVID-period effects are 0.898 in Ashanti, 0.939 in Eastern, and 0.766 in Volta, but every 95% interval includes one. The time trends are also imprecisely estimated. The fitted AR(1) coefficients range from 0.464 to 0.641, confirming appreciable serial dependence beyond the seasonal mean.

Table 3: Estimated seasonal/trend/COVID-period model with AR(1) errors by region. Amplitude is the first-harmonic semi-amplitude on the log scale; trend is the per-year log change; and the COVID effect is multiplicative on the count scale with a Wald 95% interval.

Region	R^2	Amplitude	Peak	Trend/yr	COVID effect [95% CI]	AR(1) ϕ	Harmonics $ t > 2$
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Ashanti	0.482	0.111	Jul	0.012	0.898 [0.786, 1.027]	0.464	4 of 4
Eastern	0.610	0.208	Aug	0.011	0.939 [0.799, 1.103]	0.475	3 of 4
Volta	0.659	0.341	Sep	-0.011	0.766 [0.585, 1.002]	0.641	2 of 4

5.4 State structure after seasonal AR(1) filtering

After seasonal and AR(1) filtering, the calendar concentration of decoded states becomes small (Table 4). The standard deviation of the twelve monthly higher-state shares falls from 0.231 to 0.035 in Ashanti, from 0.387 to 0.047 in Eastern, and from 0.211 to 0.035 in Volta (Figure 3, left).

Table 4: State-structure diagnostic based on innovations from the seasonal AR(1) model. “Month-share SD” is the standard deviation of the twelve monthly higher-state shares. Positive Δ BIC and Δ AIC values favor the single-Gaussian innovation model.

Region	Month-share SD (raw)	Month-share SD (innov.)	Δ BIC	Δ AIC	Innov. effect size	Innov. overlap
Ashanti	0.231	0.035	14.2	1.4	1.76	0.039
Eastern	0.387	0.047	16.7	3.8	1.46	0.153
Volta	0.211	0.035	18.4	5.6	0.56	0.222

The innovation HMM is not selected in any region (Figure 3, right). Relative to a single Gaussian, its BIC is higher by 14.2 in Ashanti, 16.7 in Eastern, and 18.4 in Volta. AIC also favors the single Gaussian in all three regions. The apparent state process in the raw monthly counts is therefore accounted for by the seasonal mean and first-order serial dependence.

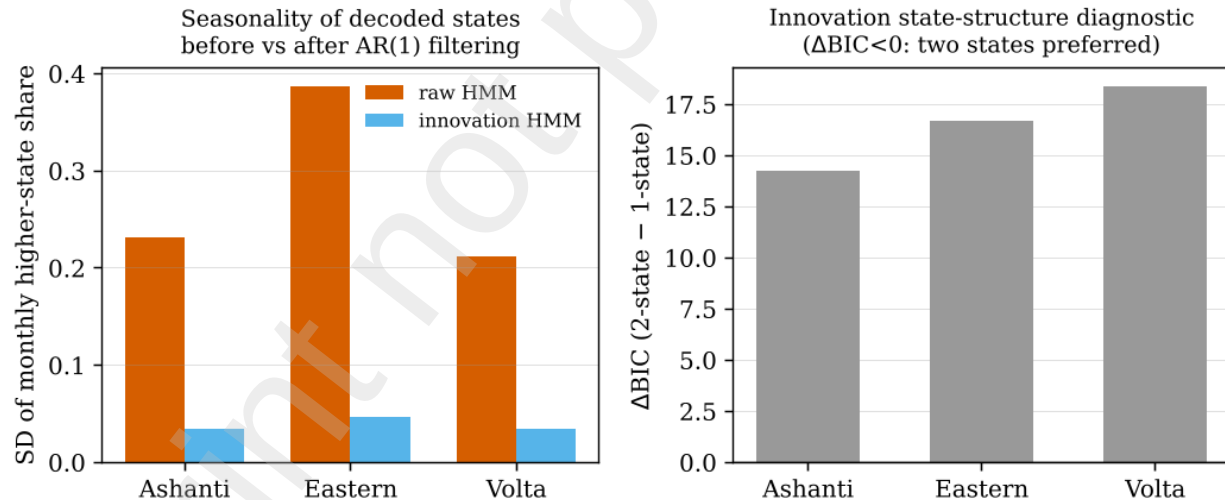


Figure 3: Innovation state-structure diagnostic. Left: the calendar dispersion of decoded higher-state shares falls after seasonal AR(1) filtering. Right: the BIC changes from replacing a single Gaussian with a two-state HMM for the innovations; positive values favor the single Gaussian in every region.

5.5 Two-state HMM summaries

Although the cross-family comparison does not select the HMMs, the two-state summaries remain informative as descriptive classifications. Table 5 reports the fitted lower and higher state count summaries, transition persistence, and emission separation. In Ashanti, the lower and

higher state medians are 69,876 and 87,074 monthly cases, with lognormal means of 70,340 and 87,532. In Eastern, the corresponding medians are 50,745 and 70,649, with lognormal means of 51,190 and 71,231. In Volta, they are 18,879 and 31,747, with lognormal means of 19,674 and 32,187.

Table 5: Two-state HMM emission and transition summaries

Region	State	Median count	Mean count	SD log count	Self-transition	Duration	Overlap
Ashanti	lower	69,876	70,340	0.115	0.880	8.31	0.311
Ashanti	higher	87,074	87,532	0.102	0.757	4.12	0.311
Eastern	lower	50,745	51,190	0.132	0.752	4.03	0.204
Eastern	higher	70,649	71,231	0.128	0.826	5.75	0.204
Volta	lower	18,879	19,674	0.287	0.870	7.67	0.239
Volta	higher	31,747	32,187	0.166	0.874	7.92	0.239

The standardized separation between lower and higher emissions is nontrivial in all three regions. The pooled-standard-deviation effect sizes are 2.021 for Ashanti, 2.542 for Eastern, and 2.215 for Volta. At the same time, the fitted normal overlap coefficients are not negligible: 0.311, 0.204, and 0.239, respectively. This combination is important. The states are not arbitrary labels, but neither are they perfectly separated classes. Months near the boundary can plausibly be assigned to either state under the fitted emission distributions.

Decoded state occupancy also differs by region. Ashanti is decoded as the higher state in 29 of 96 months, or 30.2% of the observation window. Eastern is decoded as the higher state in 54 months, or 56.3%. Volta is decoded as the higher state in 51 months, or 53.1%. These occupancy shares should not be interpreted as the proportion of months with high malaria incidence. They are model-based classifications of reported counts, and their meaning depends on the omitted seasonal and reporting structure.

5.6 Uncertainty in HMM quantities

Block-bootstrap intervals show that some HMM summaries are much less precise than their point estimates suggest (see Table A.6 in the Appendix). Higher state mean counts are relatively stable across replicates, but expected durations are much more uncertain. For Ashanti, the higher-state expected duration has a bootstrap median of 3.59 months and an approximate 95% interval from 2.25 to 17.57 months. For Eastern, the corresponding interval is 3.58 to 8.15 months. For Volta, it is 4.50 to 19.24 months.

The lower-state duration intervals are also wide: 3.59 to 17.06 months for Ashanti, 2.37 to 13.81 months for Eastern, and 3.28 to 16.07 months for Volta. These intervals make precise statements about regime persistence inappropriate. The HMMs can describe state persistence under their own assumptions, but the short monthly series and strong seasonality limit the stability of duration estimates.

5.7 Seasonality in decoded HMM states

Decoded higher states are strongly seasonal (Figure 4). Ashanti has no higher-state assignments in January, February, March, or December. Its highest monthly share is in June,

where 5 of 8 observations are decoded as higher state. Eastern is even more concentrated: January and February have no higher-state months, March has one, and June, July, and August each have higher-state shares of 1.000. Volta has a broader high-state season, with July through November each at a higher-state share of 0.750 and December at 0.500.

This pattern is the clearest substantive explanation for why the seasonal/trend/COVID-period Gaussian baseline outperforms the HMMs. The HMM states are not merely detecting irregular transitions between persistent burden regimes; they are strongly aligned with calendar month. That does not make the state labels useless. A seasonal high-count classification can still be operationally descriptive. But it means the labels should be interpreted as summaries of the annual cycle unless a model with explicit seasonal structure still supports additional latent regimes.

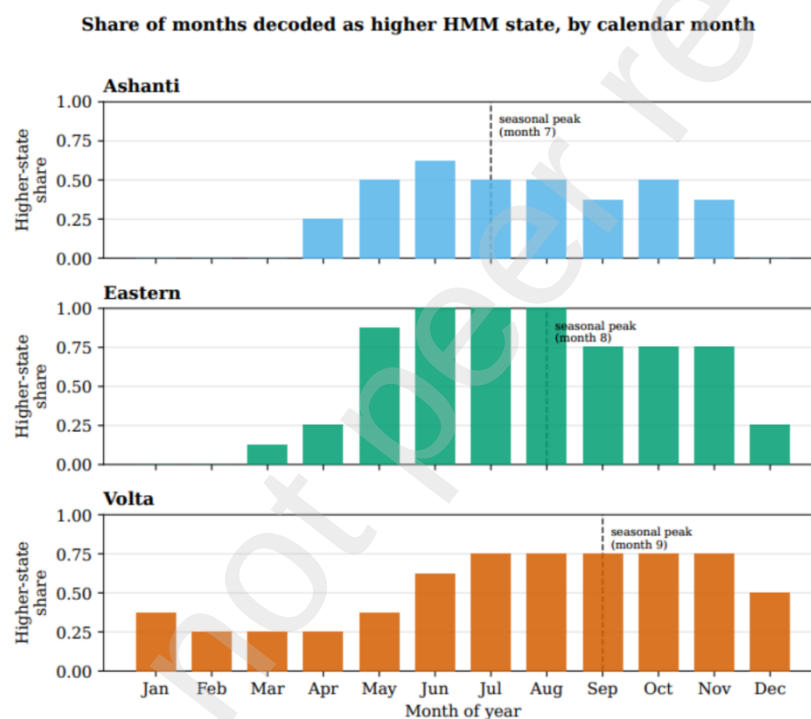


Figure 4: Share of observations decoded as higher HMM state by month of year. Each region contributes eight observations per month, one for each year from 2016 through 2023. The concentration of higher-state labels in predictable calendar months indicates that the stationary HMM is partly acting as a seasonal classifier.

5.8 COVID-period diagnostics

The supplied COVID-period indicator is associated with different decoded-state shifts across regions. In Ashanti, the higher-state share declines from 0.353 before the COVID-period indicator to 0.244 during the COVID-period indicator. In Volta, the decline is larger, from 0.765 to 0.267. Eastern moves in the opposite direction: the higher-state share increases from 0.510 before the COVID-period indicator to 0.622 during it.

The estimated COVID-period coefficients are negative on the log scale, corresponding to multiplicative effects below 1 in all three regions (0.898, 0.939, and 0.766).” The AR(1)

coefficients themselves are positive. Their confidence intervals include one, so the data do not distinguish these estimates from no COVID-period change after seasonality and serial dependence are included. The occupancy differences remain descriptive. The dataset does not contain care-seeking measures, facility-level reporting completeness, diagnostic test volumes, intervention interruptions, or alternative explanations for changes during this period.

5.9 Predictive and decoding diagnostics

PIT summaries do not show gross marginal calibration failure, but they do show residual temporal structure (see Table A.8 in the Appendix). The PIT means are close to 0.50 in all regions: 0.502 in Ashanti, 0.501 in Eastern, and 0.499 in Volta. The KS distances from uniformity are also modest: 0.053, 0.050, and 0.065. These values suggest that the fitted HMM predictive distributions are not obviously miscentered in a marginal sense.

However, the lag-1 autocorrelation in PIT values is substantial. It is 0.377 in Ashanti, 0.490 in Eastern, and 0.564 in Volta. This residual autocorrelation indicates that the HMM predictive sequence still contains temporal dependence not fully represented by the fitted model. In the context of the seasonality results, this is consistent with the need for explicit seasonal or autoregressive structure.

Posterior decoding confidence is moderate to high but not perfect. The mean posterior probability of the decoded state is 0.892 in Ashanti, 0.913 in Eastern, and 0.929 in Volta. The share of months with decoded posterior probability below 0.80 is 18.75% in Ashanti, 18.75% in Eastern, and 9.38% in Volta. These values are compatible with useful descriptive state labels, but they do not rescue the stronger interpretation that the states represent seasonally independent burden regimes.

6. Discussion

6.1 Principal finding

Simple Gaussian HMMs classify monthly malaria counts in Ashanti, Eastern, and Volta into lower and higher regimes with descriptive coherence. Seasonal models explain the raw series better, however, and the decoded states align strongly with calendar month. After the seasonal mean and AR(1) dependence are removed, the calendar concentration largely disappears and neither AIC nor BIC supports an additional two-state process. The raw-count HMM states are therefore summaries of the annual cycle and serial dependence rather than evidence of a separate latent regime process.

The state labels remain useful descriptions of lower- and higher-count months. Their meaning is narrower than a transmission-regime interpretation because the same pattern is represented more directly by seasonal and autoregressive terms.

6.2 Implications for routine malaria surveillance

For malaria surveillance programs, latent-state models are most informative when accompanied by simpler diagnostic checks rather than used as standalone classification devices. Raw-count summaries and transparent seasonal and trend baselines establish whether a more complex latent-state model improves predictive or explanatory performance. Operational

interpretation of state labels also requires checking their alignment with calendar month, reporting periods, and known intervention cycles.

The distinction between reported count regimes and incidence regimes is also important. Without denominators, a region with more reported cases may have higher population, higher facility attendance, better reporting completeness, or higher incidence. Similarly, a transition into a higher reported-count state may reflect true transmission increases, but it may also reflect improved reporting or changes in care seeking. For policy interpretation, the stronger claim requires population at risk, testing volume, reporting completeness, or facility-level context.

6.3 Practical reporting recommendations

Routine-surveillance HMM studies should identify whether the modelled quantity is a count, rate, test positivity measure, or another surveillance outcome. A count-regime model does not represent incidence regimes unless denominators and the observation process support that interpretation. At least one simple seasonal benchmark is also needed because annual seasonality is part of the malaria data-generating process. State-count assessment should include both HMM-only and cross-family comparisons; reporting only the preferred HMM can conceal a better-fitting simpler model.

Fourth, decoded state sequences should be summarized by calendar month, not only by time-series plots. A plot can make a state sequence look dynamic, while a month-of-year table may reveal that the same state appears predictably each year. Fifth, duration claims should be accompanied by uncertainty intervals. Expected duration is one of the most attractive HMM summaries for surveillance, but it can be unstable in short monthly panels. Sixth, authors should describe the COVID-period or other reporting-period variables as observational covariates unless the design can support causal interpretation.

These recommendations are not specific to Ghana. They apply whenever routine infectious disease data are used to infer latent states from aggregated monthly counts. The broader point is that interpretability should be earned by model comparison and diagnostics, not assumed because the state labels are intuitive.

6.4 Implications for spatial and spatio-temporal epidemiology

The paper also speaks to a broader issue in spatial and spatio-temporal epidemiology: interpretable labels must be connected to the observation process. Space-time clusters, latent states, and regional risk classifications are valuable when they reveal structure not already explained by design, reporting, or calendar effects. They are less useful when they relabel expected variation. The literature on space-time cluster detection and surveillance-data bias correction makes this point in different ways (Lan and Delmelle 2023; Li et al. 2020). This study provides a concrete malaria example at a smaller spatial scale.

The three-region design also clarifies what a stronger spatial paper would require. A national Ghana analysis could model all regions or districts jointly, include spatially structured random effects, use population denominators, and allow seasonal effects to vary across ecological zones. Such a model might identify state structure that is not present in these three aggregated series. The present analysis shows that the region-specific raw-count HMM does not isolate a regime beyond the observed temporal pattern.

6.5 Small-sample model selection

With 96 monthly observations per region, a three-state HMM has enough flexibility to improve likelihood. AIC rewards that improvement, whereas BIC penalizes it more strongly. Neither criterion alone establishes whether the states are epidemiologically meaningful. Comparison with the seasonal baseline shows that a transparent calendar model explains the series better; reporting only the BIC-selected two-state HMM would conceal both that result and the AIC–BIC disagreement.

6.6 Limitations

The analysis has several limitations. Provenance is limited to the public repository record. The original extraction procedure, DHIMS II indicator definitions, reporting completeness, and retrospective revisions could not be independently verified against the underlying GHS database. The findings therefore characterize the deposited dataset and should not be interpreted as official regional estimates. The outcome is reported malaria cases rather than incidence, and the absence of population denominators prevents calculation of rates and limits comparisons across regions. The dataset contains only three regions and cannot support national, district-level, or spatial diffusion claims. The region-specific models do not include spatial dependence. The COVID-period variable is descriptive rather than causal. The HMMs also use Gaussian emissions on log counts rather than a discrete count model such as Poisson or negative binomial.

Additional limitations concern the model class. The Gaussian-on-log emissions approximate a count process; the lognormal back-transformation used throughout ($\exp(\mu + \sigma^2/2)$ for the mean) corrects the obvious Jensen bias, but a negative-binomial HMM remains the natural count-based alternative and could behave differently in the low-volume Volta series. The transition matrices are time homogeneous, so the HMMs do not allow transition probabilities to vary by month, rainfall, intervention timing, or reporting conditions. A non-homogeneous HMM with seasonal transition probabilities, or a single-stage Markov-switching regression with shared harmonic terms, could represent the seasonal mean, dependence, and latent states jointly. The innovation diagnostic is two-stage and does not propagate uncertainty from the fitted seasonal AR(1) model. A seasonal state-space model or hierarchical spatio-temporal model could better separate latent malaria risk from reporting noise. Those extensions require more data and stronger assumptions than are appropriate for the current three-region monthly panel.

The bootstrap uncertainty assessment is also limited. Fifty circular block-bootstrap replicates expose instability in duration estimates but do not constitute a full uncertainty analysis. A larger methods study could use more replicates, alternative block lengths, Bayesian posterior inference, or simulation-based calibration. Here, the bootstrap is a practical robustness check.

6.7 Future work

Population denominators would allow incidence-rate modelling. Facility reporting completeness, testing volume, and outpatient attendance would help separate malaria burden from health-system observation processes, while rainfall, temperature, vegetation, and intervention variables would support more direct seasonal and ecological interpretation.

A district-level or all-region Ghana panel would support hierarchical spatio-temporal models, spatial smoothing, and region-specific seasonal profiles. Such a model could test whether

residual high-risk regimes remain after accounting for shared national trends and local seasonality, and whether Ashanti, Eastern, and Volta behave similarly to other Ghanaian regions.

Negative binomial HMMs, non-homogeneous HMMs with seasonal transition probabilities, autoregressive state-space models, and Bayesian hierarchical models are plausible alternatives. The surveillance question should determine the choice. Operational classification prioritizes interpretability and calibration, whereas inference about latent transmission requires denominators and an explicit observation process.

7. Conclusion

The HMMs produce interpretable descriptive state summaries, but the seasonal AR(1) model has lower AIC and BIC in Ashanti, Eastern, and Volta. Seasonal peaks occur from July through September, consistent with Ghana's wet-season transmission. The COVID-period coefficients are negative but imprecise after serial dependence is included. Once the seasonal mean and AR(1) dependence are removed, neither AIC nor BIC supports an additional two-state process. COVID-period occupancy shifts are region specific, and duration estimates have wide bootstrap intervals. A stationary HMM on raw monthly counts therefore summarizes seasonality and autocorrelation rather than a distinct latent regime process.

Ethics approval

This study uses publicly released, aggregate monthly regional surveillance counts and does not analyze individual-level human-subject data.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data and code availability

The secondary dataset analysed in this study is publicly available as *Indicators Data* on Mendeley Data under a CC BY 4.0 licence (Seidu, 2026). It is a depositor-provided extract and not an official GHS release.

Declaration of Competing Interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used OpenAI Codex/ChatGPT to assist with language editing, consistency checks, and reproducibility review. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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